COMPUTER VISION & BIOMETRICS PROJECT

Face Recognition Attendance System

Real-Time Facial Detection, Automated Biometric Logging & Analytical Reporting.

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PROJECT CONTEXT

Transforming Attendance Verification

Traditional Weaknesses

Proxy Marking: Manual sign-ins enable buddy punching.

Hardware Clutter: Expensive biometric scanners fail on dirty sensors.

Slow Throughput: Manual queues delay office check-ins.

The AI Solution

Contactless Sensing: Instant visual facial verification.

Zero Duplicate Logs: Enforces single-entry constraints per day.

Live Cloud/Web UI: Centralized web dashboard with CSV exports.

SYSTEM DESIGN

End-to-End Operational Pipeline

01

Staff Entry

Profile metadata stored in SQLite schema.

02

Face Sampling

OpenCV extracts 60 grayscale face frames.

03

LBPH Training

Compiles feature vectors to trainer.yml.

04

Live Matching

Camera detects face and calculates confidence.

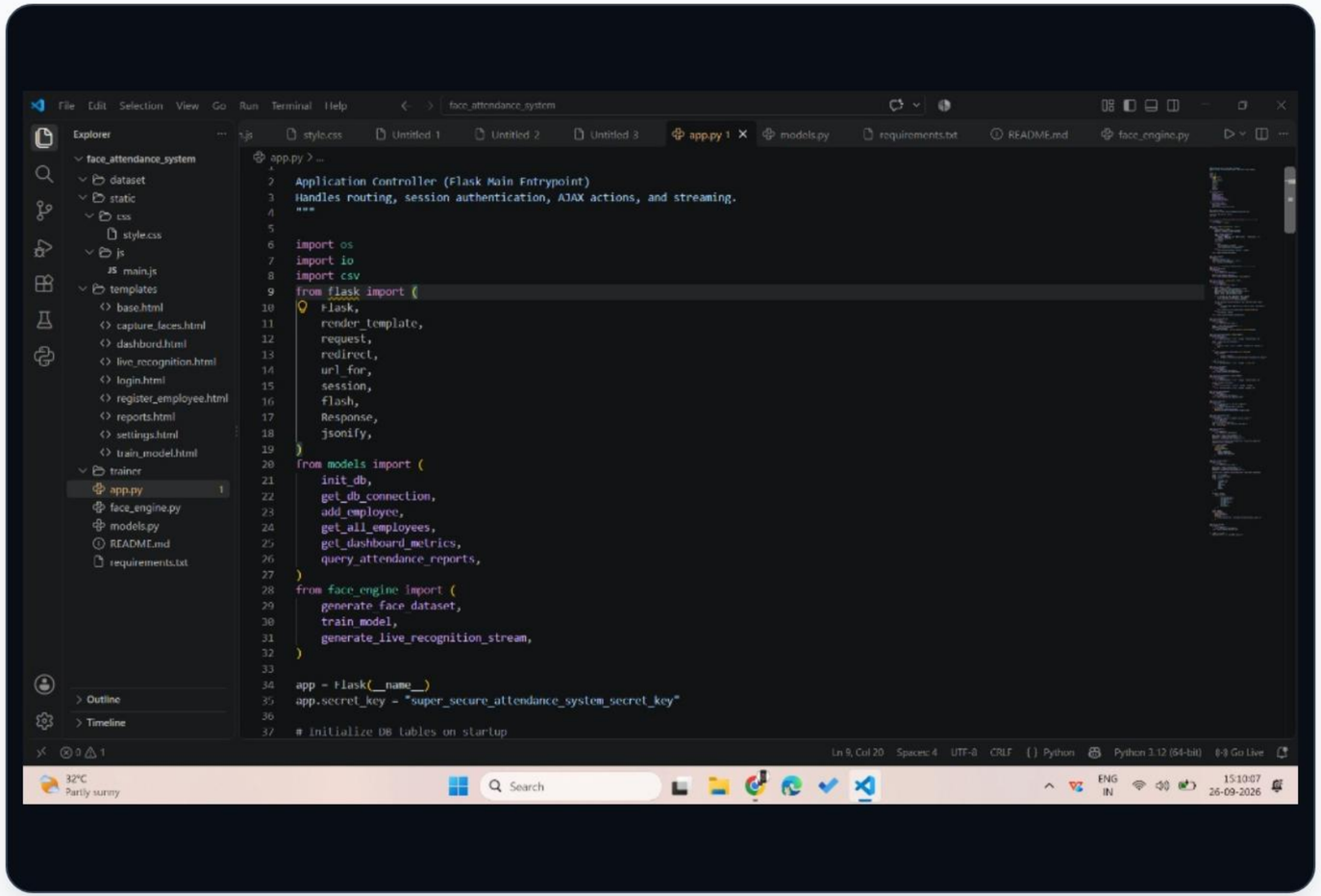
05

Auto Logging

Inserts date/time stamp without duplicates.

BACKEND IMPLEMENTATION

Flask MVC Application Architecture



Modular MVC Organization

Routing logic decoupled into clean controllers and models.

REST & Streaming Endpoints

Delivers MJPEG multipart video frames directly to browsers.

Session-Protected Routes

Restricts employee enrollment and models to admins.

ENVIRONMENT & DEPENDENCIES

Computer Vision Environment Setup

opencv-contrib-python

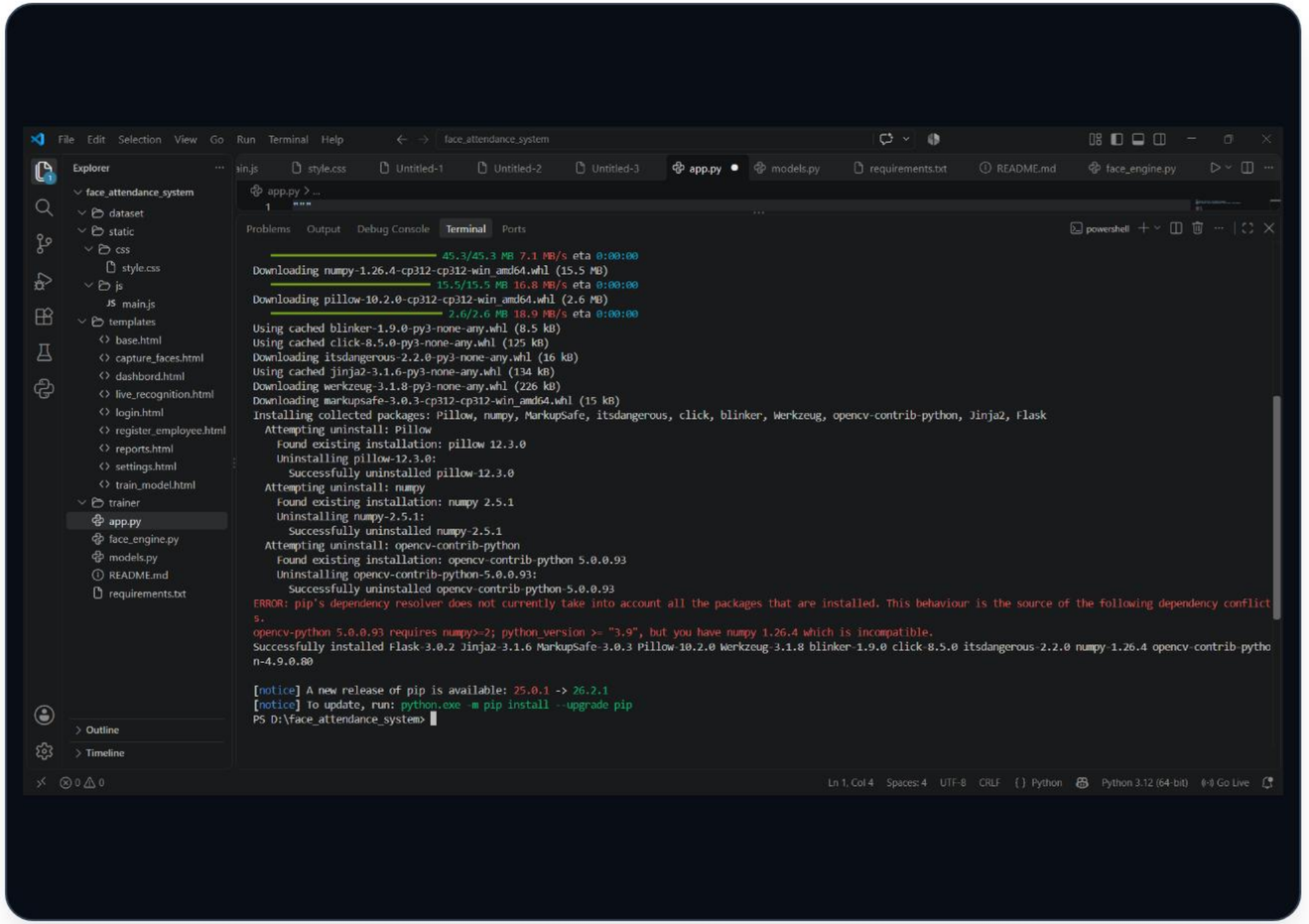
Essential module containing `cv2.face.LBPHFaceRecognizer`.

NumPy Matrix Processing

Converts captured face regions into vectorized array inputs.

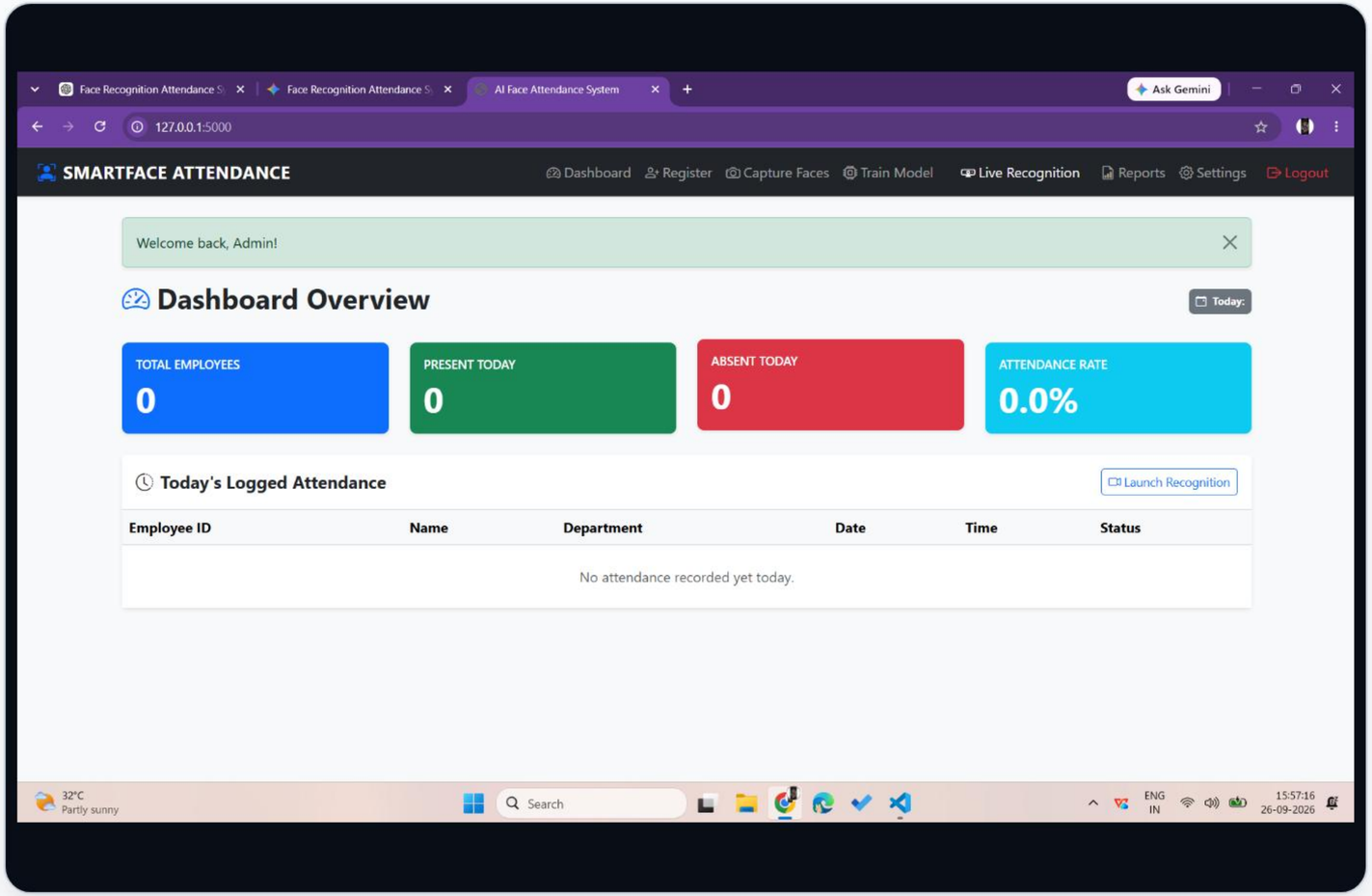
Pillow & Haarcascade

Accurate grayscale conversion and frontal face cropping.



USER INTERFACE

Real-Time Operations Dashboard



Dynamic KPI Cards

Instant metrics on total staff, present counts, and ratios.

AJAX Live Polling

Updates table rows immediately as faces are identified.

Instant Launch Hub

One-click redirect to camera streams and reports.

SYSTEM EXECUTION

Live Streaming & Automated Dataset Creation

Dataset Generation

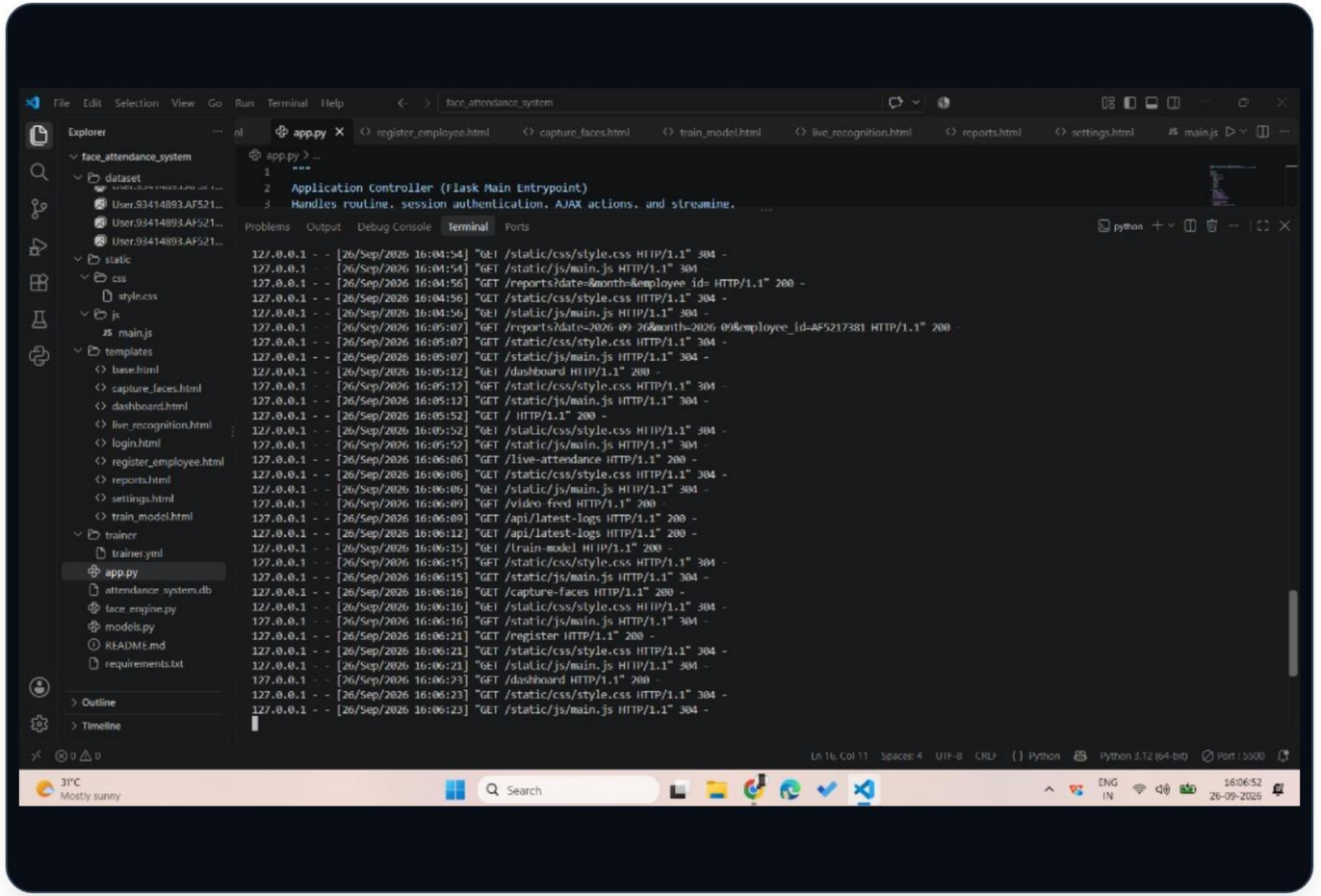
Captures 60 face frames per employee with auto-naming.

Model Weight Export

Outputs trained matrix directly to `trainer/trainer.yml`.

200 OK Telemetry

Persistent streaming logs verified across all API endpoints.



ALGORITHM MECHANICS

Haar Cascade vs. LBPH Recognizer

Haar Cascade Detector

Identifies facial region of interest (ROI) in webcam feed.

Scale Factor: 1.2 to handle depth movement.

Min Neighbors: 5 to prevent false bounding boxes.

Output: Bounding box coordinates (x, y, w, h).

LBPH Recognizer

Local Binary Patterns Histograms for identity matching.

Lighting Robust: Evaluates local micro-patterns.

Distance Scoring: Distance < 75 triggers identity match.

Efficiency: Real-time performance without heavy GPUs.

DATA PERSISTENCE

Relational Schema & Integrity Rules

Employees Table

Column	Type	Constraint
employee_id	TEXT	PRIMARY KEY
name	TEXT	NOT NULL
department	TEXT	NOT NULL
email	TEXT	UNIQUE

Attendance Table

Column	Type	Constraint
attendance_id	INTEGER	AUTO INCREMENT
employee_id	TEXT	FOREIGN KEY
date, time	TEXT	NOT NULL
Constraint	**UNIQUE(employee_id, date)**	

SYSTEM PERFORMANCE

Empirical Benchmarks & Speed

<0.5s

Recognition Latency

Near-instantaneous recognition and DB write.

60

Face Frames

Samples per person for optimal variation.

100%

Duplicate Prevention

Zero multi-entry glitches per working day.

ROADMAP

Future Scalability & Roadmap

Anti-Spoofing

Blink detection and infrared checks to counter photo spoofing.

Deep Learning

Migrating to FaceNet embeddings for enterprise-scale matching.

PostgreSQL Sync

Centralized cloud cluster for multi-branch attendance sync.

Thank You

Face Recognition Attendance System with Real-Time Detection and Automated Attendance Logging.

Questions & Answers

Open for Live Demo